



Princeton Computer Science Contest – Fall 2025

Problem 8: Fall Foliage (20 points) [File Upload]

By Frederick Qiu

The date is October 26th, and the trees are all shades of red and orange on campus. Your friend is visiting for the weekend, and is about to catch the train to go back home. Before they leave, they want to see all the fall foliage on campus and take some nice photos. Unfortunately, the train is arriving soon so time is short. The formal setup is as follows:

- n^2 trees are arranged in an $n \times n$ grid, where n is even. The trees have a fixed but unknown ranking.
- You may start from any tree on the edge of the grid and move up, down, left, or right.
- When you visit a tree, you learn whether that tree is the best tree visited so far.
- When you visit a tree, you may take photos at the tree. You can only do this once.
- You must visit every tree exactly once and end adjacent to your starting point.

Part 1 (10 points)

Give a randomized algorithm that takes photos at the best tree with probability $\Omega(1/n)$, and prove your algorithm is correct. You need not rigorously describe your algorithm so long as it is clearly conveyed.

Part 2 (10 points)

Due to the roads around campus, it turns out that the best tree for a photo shoot will definitely not be on the edge of the $n \times n$ grid. With this extra assumption, give a randomized algorithm that takes photos at the best tree with probability $\Omega(1)$, and prove your algorithm is correct. As in Part 1, you need not rigorously describe your algorithm.

Background

This problem is an example of an *optimal stopping problem*, where we observe a sequence of choices and try to “stop” at the best option. They model many real-world scenarios, such as deciding whether to accept a job offer or continue searching. If you’re stuck, come to office hours or read more about optimal stopping problems [here](#)!

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